

Fully Automated Software-Defined Switched Quantum Key Distribution Network Enhanced with PUF Authentication

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Abstract We demonstrate a fully automated 4-node switched-QKD network using PUF-based authentication orchestrated by a central SDN controller, enabling scalable and dynamic operation. ©2026 The Author(s)

Introduction

The rapid development of quantum computing has raised new concerns about the robustness of classical cryptographic frameworks. To ensure long-term security, Quantum Key Distribution (QKD) [1] has been proven to be a vital solution, exploiting the principles of quantum mechanics to provide information-theoretically secure (ITS) keys. While early QKD deployments were mainly point-to-point, modern network requirements demand dynamic, scalable configurations capable of all-to-all (every Alice to every Bob) secure communication. Recent research has demonstrated that switched-QKD, utilizing low-loss optical switches (LLOS), can significantly improve resource efficiency, offering savings of up to 28% in large-scale deployments [2],[3]. Also, with the integration of Software Defined Networking (SDN), centralized control has been enabled in switched-QKD networks [4]. Despite these advances, a critical bottleneck in QKD networks still remains: initial authentication.

Currently, most QKD networks rely on vendor-specific authentication schemes using Pre-Shared Keys (PSKs). This static approach possesses security risks and is inherently unscalable, since a network of N QKD pairs requires $N(N-1)/2$ manually managed PSK pairs to achieve full connectivity, as depicted in Fig. 1a. To address this, Physical Unclonable Function (PUF)-based authentication (PBA) offers a hardware-intrinsic and scalable alternative. PUFs exploit random microscopic manufacturing variations of a component to derive unique device-specific keys or responses, whose security relies on the physical unclonability of the hardware [5]. While previous works have explored PBA in isolated contexts [6]-[10], they lack integration into a fully dynamic, switched environment with dedicated PUF authenticators for each QKD module.

While our previous works [11],[12] established the baseline for SRAM PUF-based dynamic authentication with preliminary switch management,

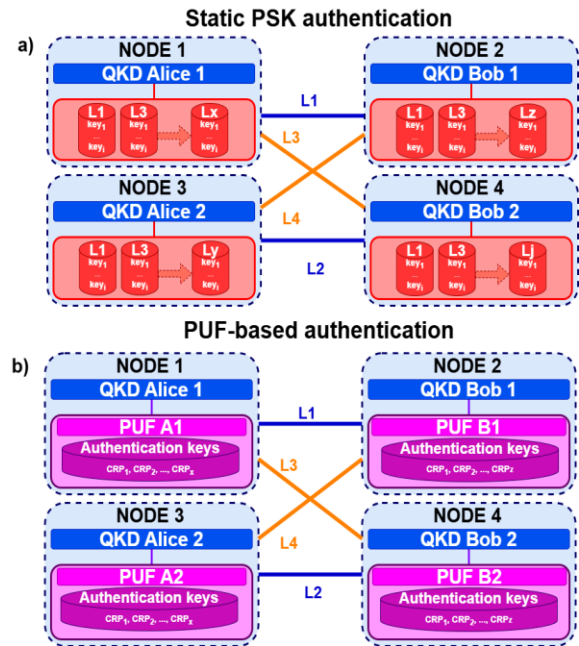


Fig. 1: (a) In static PSK implementation each link requires one dedicated database for PSK storing, while in (b) every node necessitates only one PUF authenticator.

this work focuses on the operational requirements of a fully automated switched-QKD network. To achieve this, we extend that architecture by introducing a centralized SDN controller that continuously monitors network conditions and dynamically orchestrates switching operations based on buffer occupancy and key consumption. The enhanced system enables adaptive, demand-driven reconfiguration, allowing uninterrupted operation over 20 hours while maintaining quantum-safe key availability for all links.

Experimental setup

The experimental testbed consists of a four-node switched-QKD network with initial PBA controlled by a central SDN server, as illustrated in Fig. 2. The setup includes two QKD pairs implementing the BB84 protocol with polarization-based encoding (ThinkQuantum Qukey) and four PUF-based authenticators developed by Qubitech. The QKD

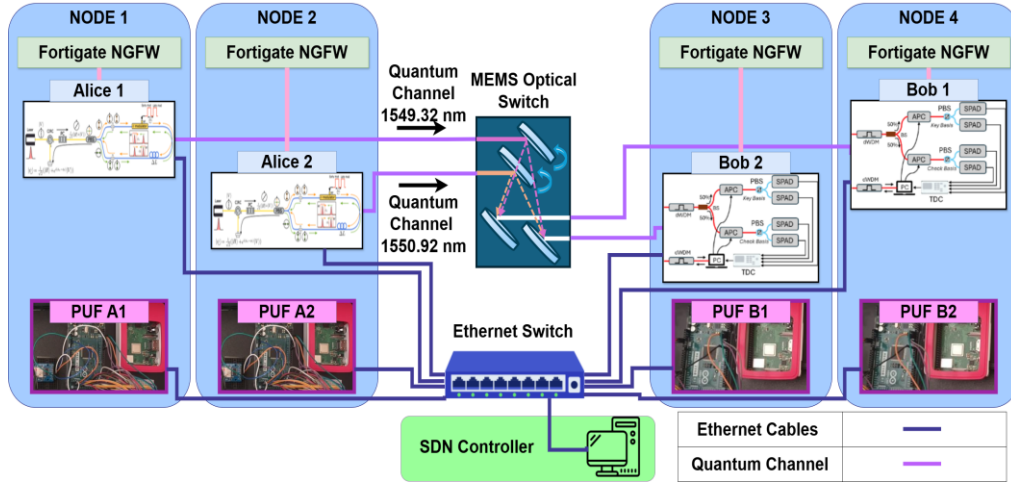


Fig. 2: The 4-node network schematic of the QKD modules with the PUF-based authenticators orchestrated by the SDN controller. The quantum channels are connected to a central LLOS enabling operation either in BAR or CROSS configuration.

devices employ the iPOGNIAC encoder [13], enabling qubit synchronization through the quantum channel, operating at 1549.32 nm (ITU Channel 35) and 1550.92 nm (ITU Channel 33). The QKD systems employed support two key functionalities essential for this demonstration: i) on-demand configuration of initial authentication keys, and ii) continuous operation of the Key Management System (KMS), which provides QKD keys across all links between switching operations. The network operates under two configurations: the BAR configuration, where Alice 1 connects to Bob 1 and Alice 2 to Bob 2, and the CROSS configuration, where Alice 1 connects to Bob 2 and Alice 2 to Bob 1. Accordingly, the links are denoted as L1 and L2 for the BAR configuration, and L3 and L4 for the CROSS configuration as shown in Fig. 1. Key buffers that are associated with active links are defined as ACTIVE, while the remaining ones are considered PASSIVE. The SDN orchestrator is realized through a python-based routine that uses open-source libraries to orchestrate all testbed operations. For authentication, each QKD device is equipped with an SRAM-PUF module, that generates ephemeral, unique keys for every link following each switching operation. The PUF instances are implemented on a RISC-based microcontroller interfacing with a Cypress CY62256NLL-70PC SRAM unit, while an ARM-based SoC manages challenge-response pairs and key validation during enrolment and authentication phases. At the application layer, a quantum-safe video conferencing service was deployed over a QKD-enhanced IPsec channel established using four Fortigate NGFW encryptors.

Automated Control

To enable dynamic control of the network, a dedicated monitoring process is instantiated during initialization. This process continuously monitors the KMS upon its buffer levels and key material consumption rates. Based on the observed key

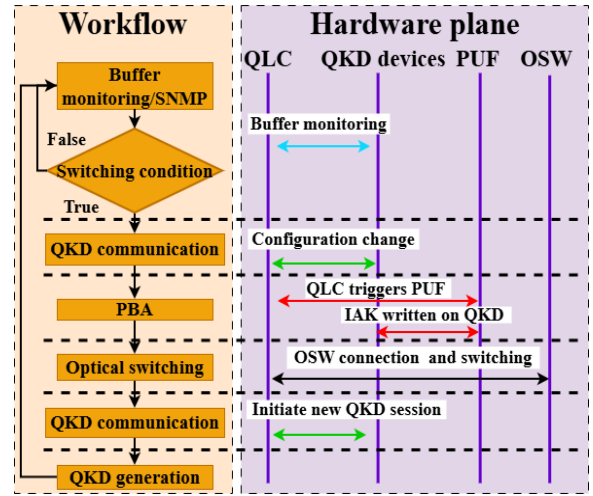


Fig. 3: Workflow schematic of the executed processes (request and response) during switching operations.

consumption rates, the SDN controller classifies buffers into priority and non-priority categories. Priority is enforced by applying more flexible threshold conditions for switching between BAR and CROSS configurations. As consumption rates change, these priorities are dynamically updated to reflect real-time network conditions. The monitoring and decision-making algorithm aims to maintain balanced buffer levels while preventing both depletion and excessive imbalance across buffers. To describe threshold levels and switching criteria the following parameters are defined. C , x_p , x_n , T_U , T_L , Δ_U , Δ_L . where C is the buffer capacity in bytes, x_p , x_n are the buffer occupancy values for priority and non-priority buffers. T_U , T_L , are the upper and lower threshold levels for the priority buffer while Δ_U , Δ_L are the lower and upper imbalance thresholds. For a switching operation to occur two CONDITIONS are checked. **Condition 1 (C1):** $x_p > T_U$ AND $x_p - x_n > \Delta_U$ and **Condition 2 (C2):** $x_p < T_L$ OR $x_p - x_n < \Delta_L$. For simplicity C1 denotes where non-priority buffers require replenishment, whereas C2 the need for priority buffers to be refilled. When

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